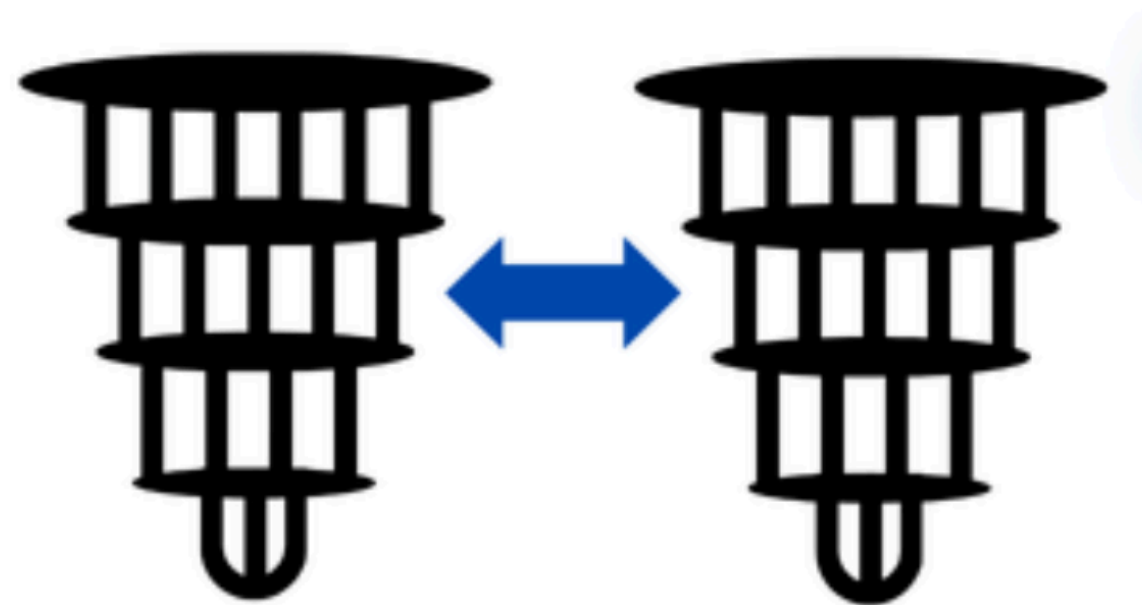


Distribuyendo Computación Cuántica, la próxima frontera

Una introducción a los aspectos teóricos de la computación cuántica distribuida, destacando sus fundamentos, retos y oportunidades para el futuro de la computación cuántica.

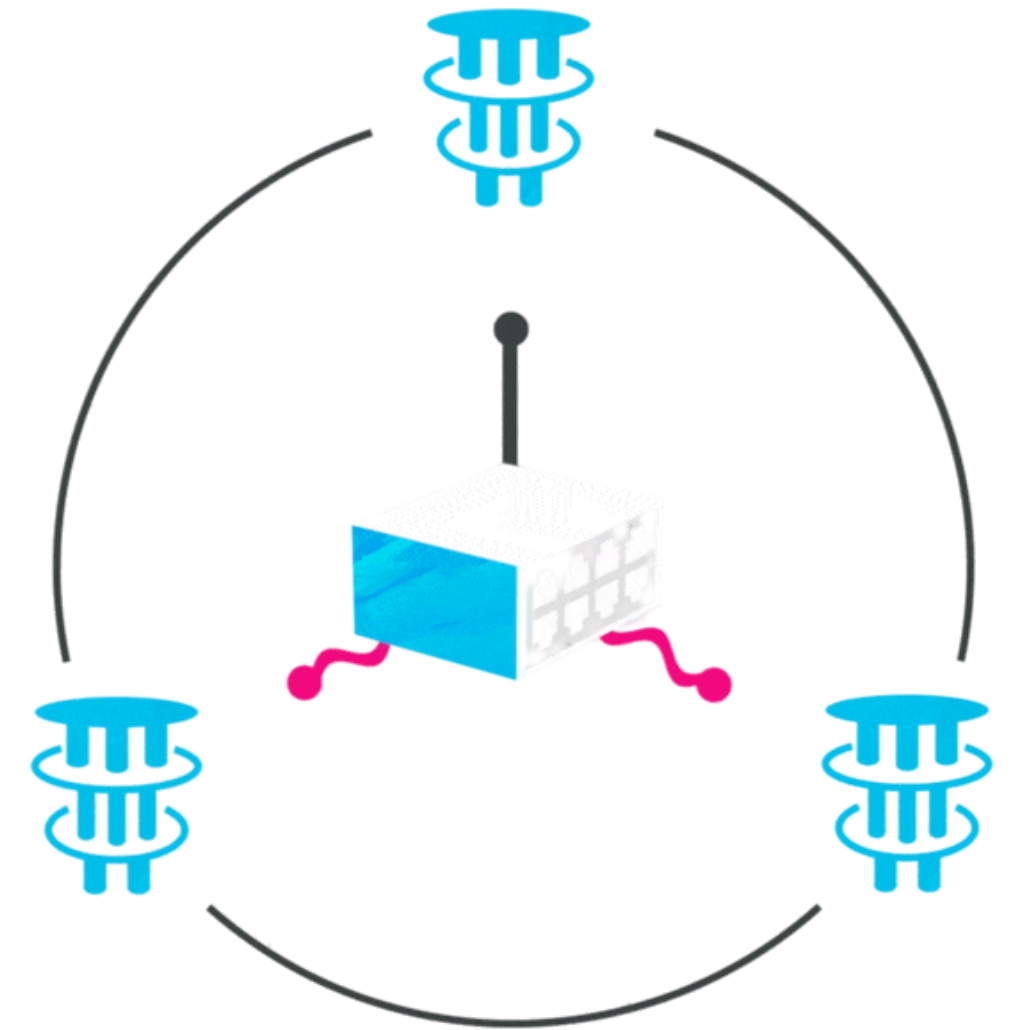


María Gragera Garcés



¿Por qué distribuir la computación cuántica?

- Los ordenadores cuánticos actuales tienen un número **limitado** de qubits y fidelidad.
- Aumentar el tamaño de un solo dispositivo es técnicamente complejo y costoso.
- La distribución cuántica busca **interconectar** múltiples procesadores para **cooperar** en un mismo cálculo.
- Esto permitiría **escalar** de forma modular, reutilizar hardware existente y acercar la era del cómputo cuántico útil.



¿Qué es la distribución cuántica?

Paradigma donde varios ordenadores cuánticos se conectan y cooperan para ejecutar un programa conjunto.

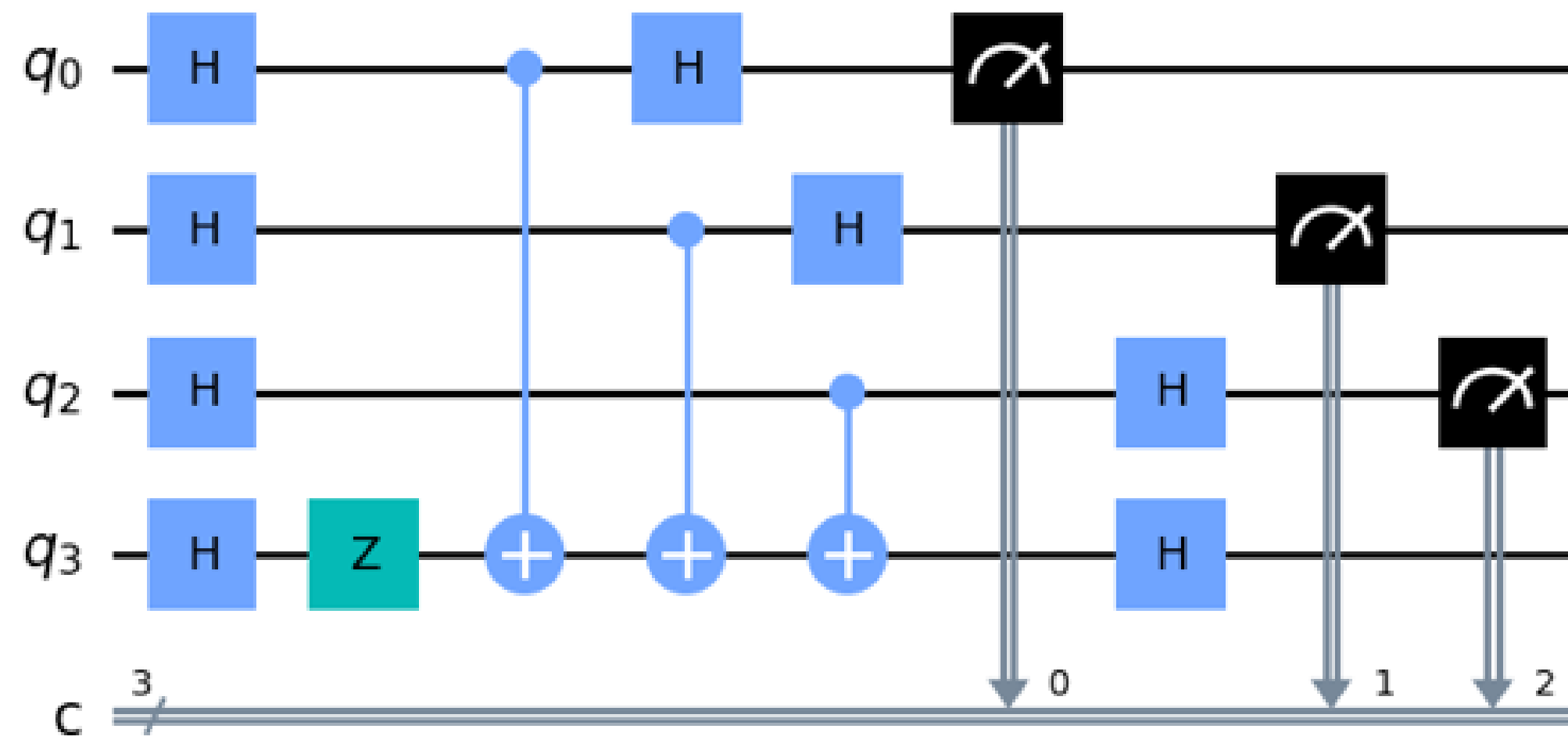


- Cada nodo mantiene su propio conjunto de qubits locales y se comunica con otros a través de enlaces cuánticos.
- A diferencia de la computación clásica distribuida, la comunicación debe preservar entrelazamiento y coherencia cuántica.

Operaciones distribuidas

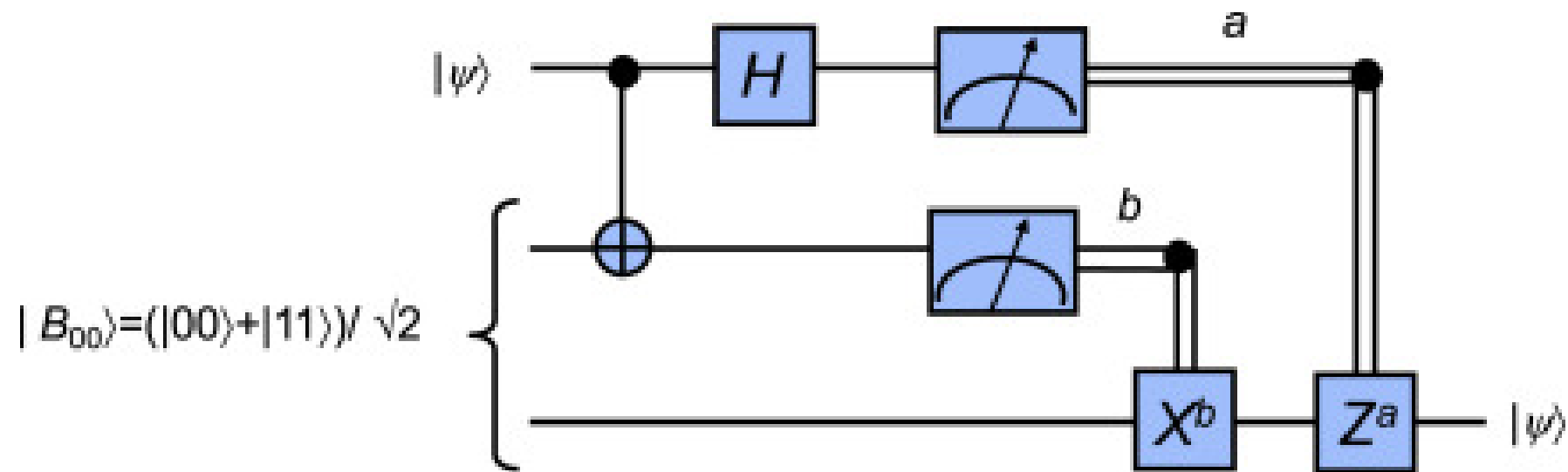
Teletransportación

Operaciones no locales



Operaciones distribuidas

Teletransportación



Operaciones no locales

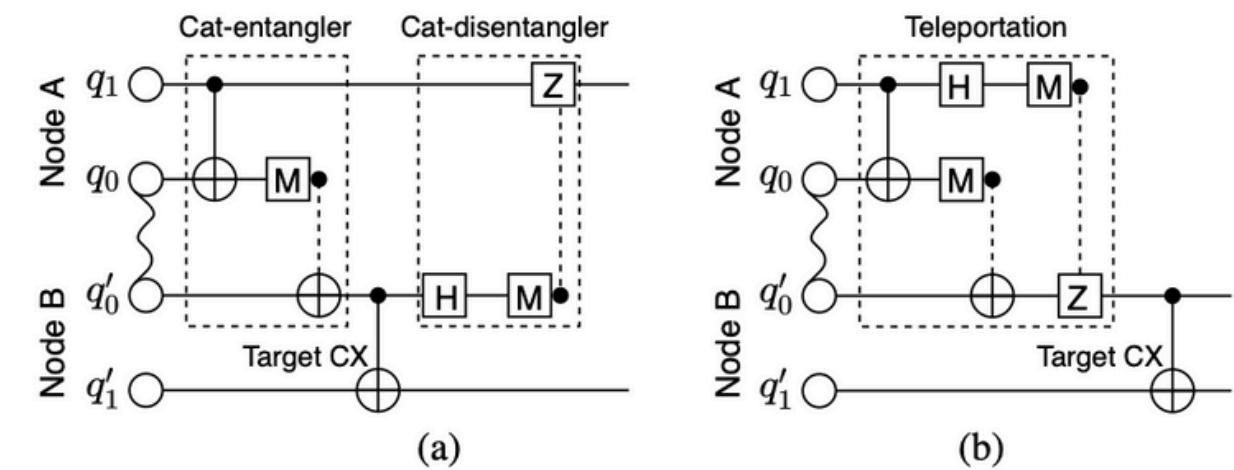
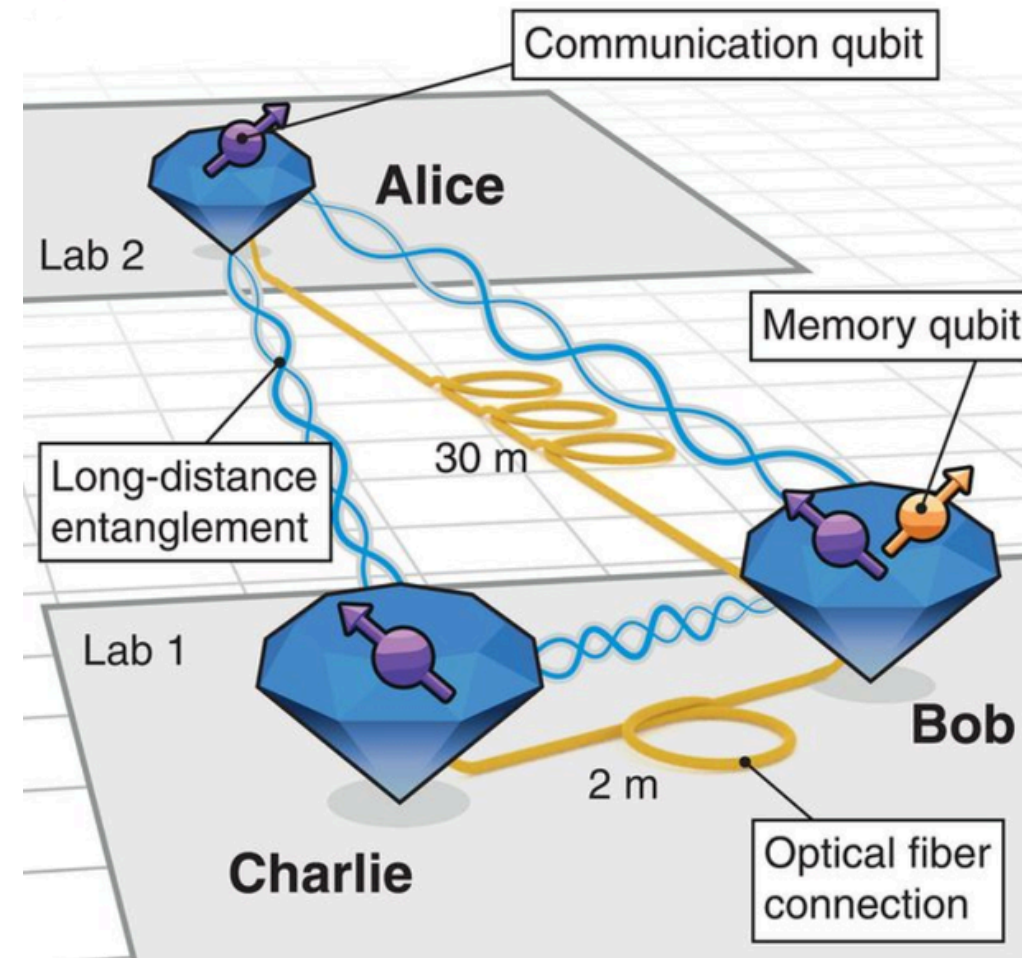


Figure 2: The implementation of one remote CX. (a) The Cat-Comm version. (b) The TP-Comm version. Each wavy line denotes an EPR pair between qubits, and each dashed line denotes one bit of classical communication. M denotes measurement.

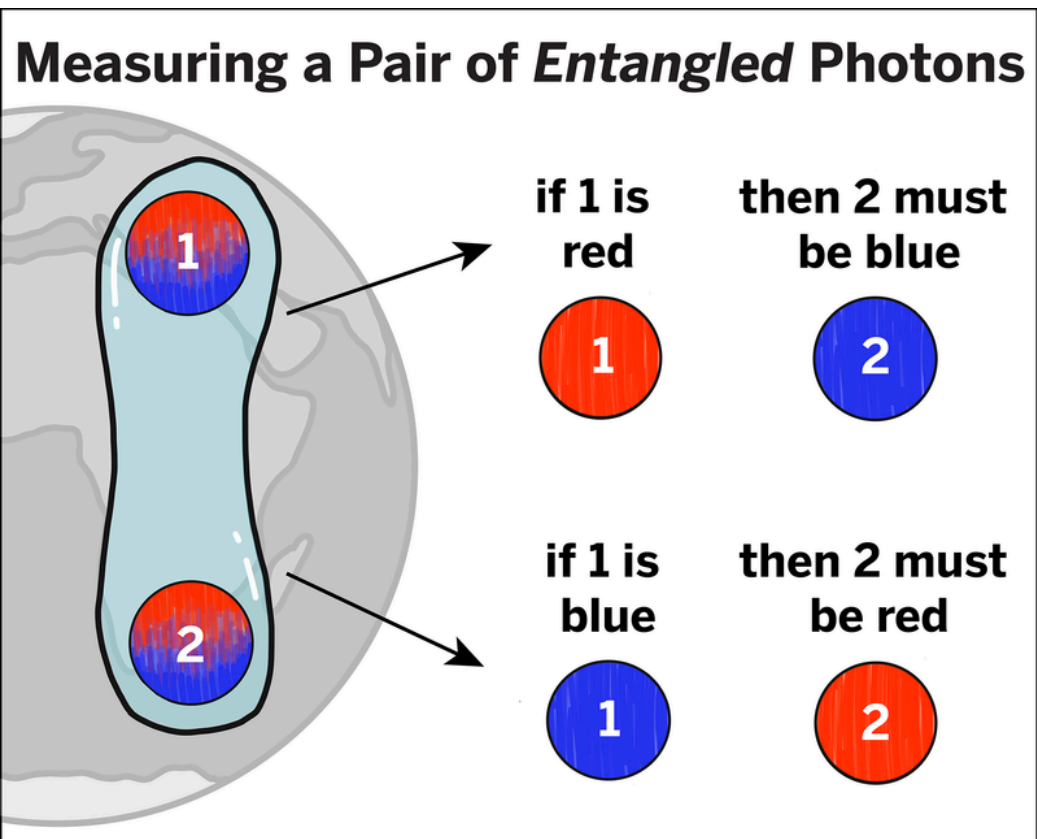
Redes cuánticas



El propósito de una red cuántica es distribuir entrelazamiento entre nodos remotos, habilitando tareas como:

1. Teletransportación cuántica de información entre procesadores.
2. Sincronización y comunicación seguras mediante claves cuánticas.
3. Computación cuántica distribuida, donde varios nodos cooperan en una misma tarea.

Teletransportación cuántica



La teletransportación cuántica permite transferir el estado cuántico de una partícula a otra sin mover físicamente el qubit.

Funciona gracias al entrelazamiento entre dos partículas compartidas y a un canal clásico que transmite los resultados de una medición.

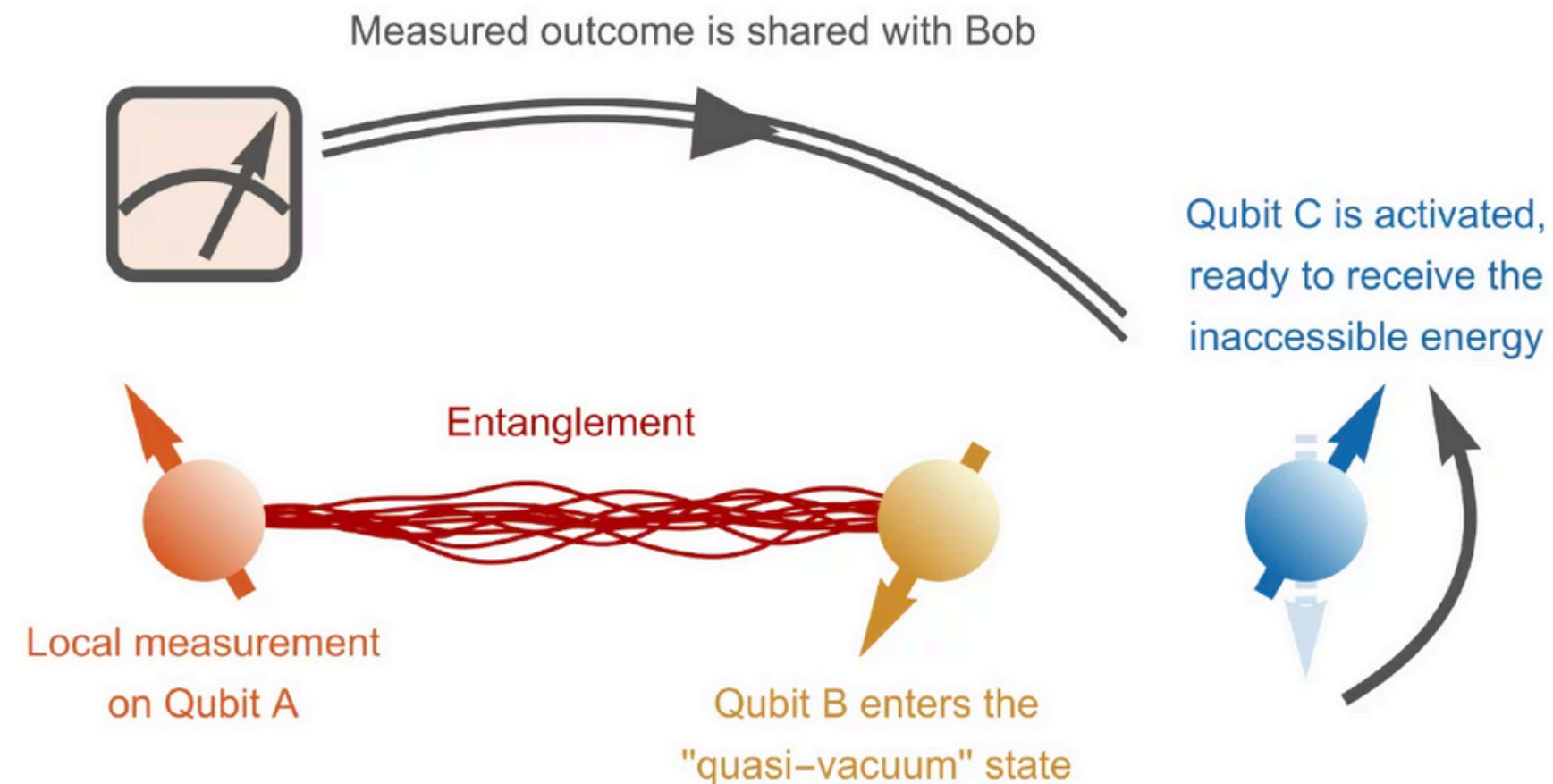
El proceso requiere tres elementos:

1. Entrelazamiento compartido entre el emisor (Alice) y el receptor (Bob).

2. Medición conjunta del qubit original y la mitad del par entrelazado.

3. Comunicación clásica para que Bob aplique una corrección y recree el estado original.

Así, la información cuántica se “mueve” sin viajar físicamente, manteniendo la coherencia del sistema.



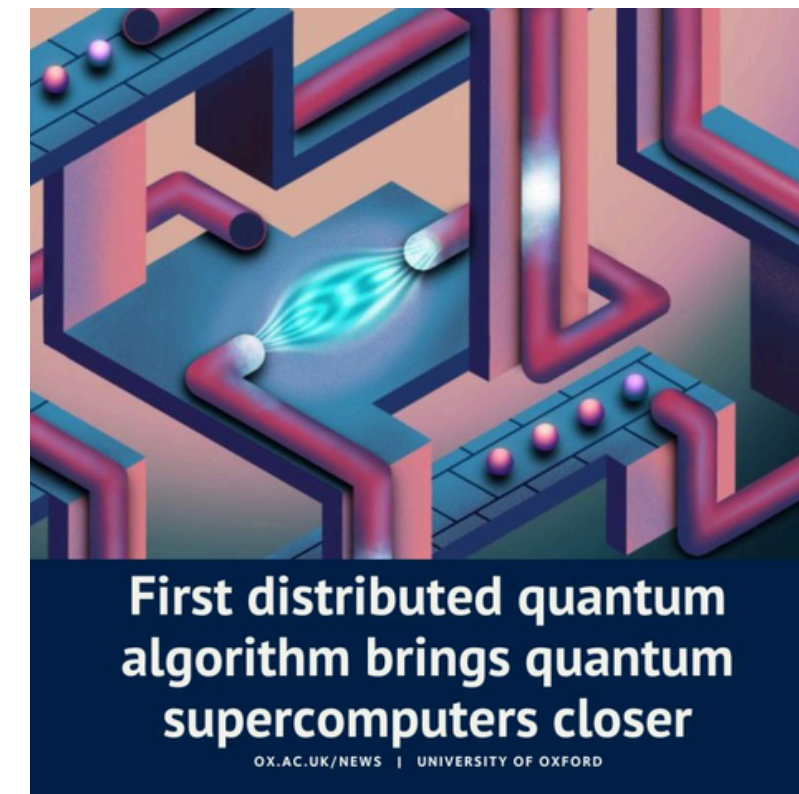
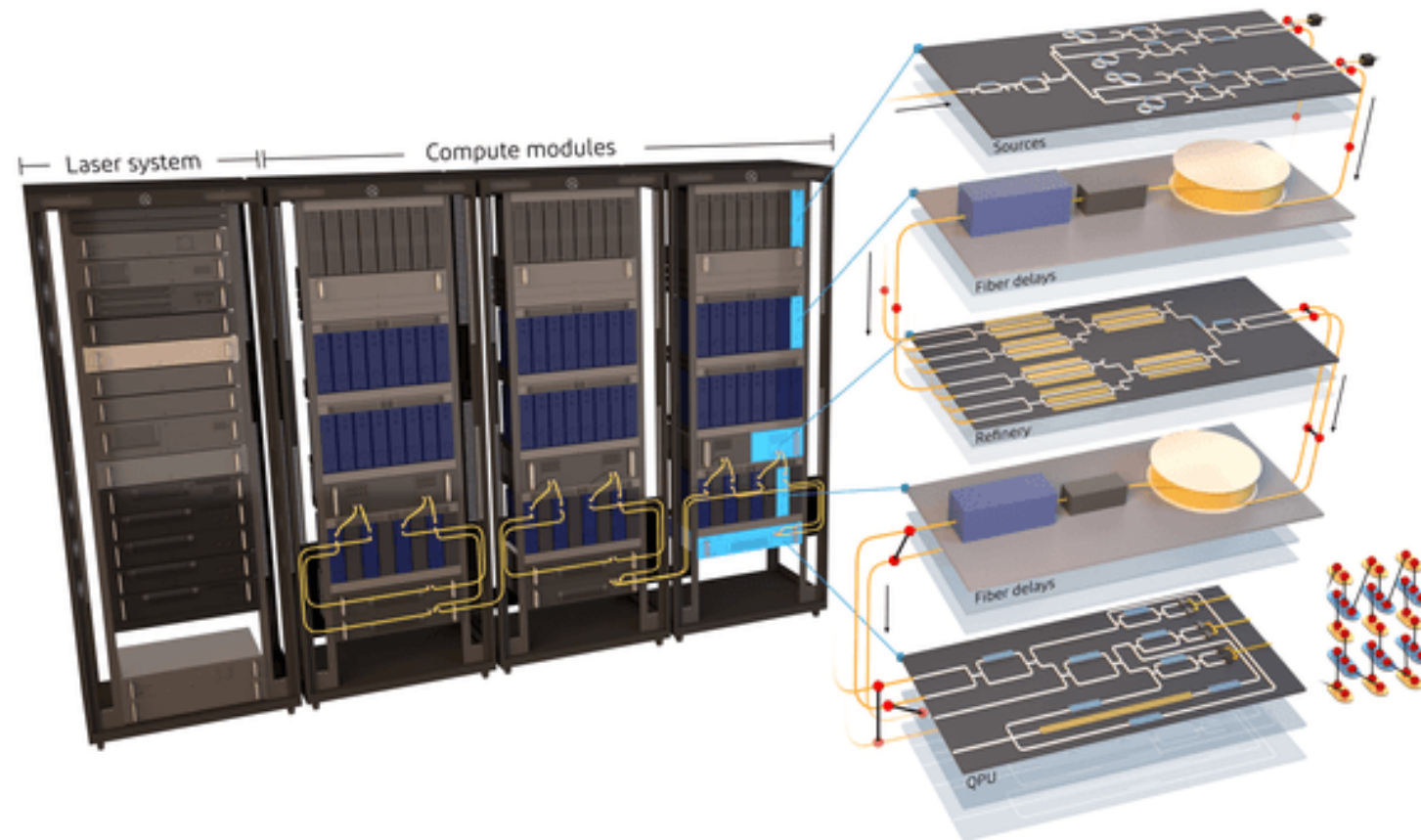
Estado del arte de DQC

La computación cuántica distribuida es un campo muy joven.

Los primeros trabajos teóricos aparecieron alrededor de 2018, y hoy ya existen demostraciones experimentales de distribución.

- Xanadu anunció hace un año Aurora, el primer ordenador cuántico distribuido (aun no es publico).
- Un grupo en la universidad d Oxford también logró una teletransportación entre dos procesadores, un paso modesto pero significativo hacia la computación distribuida real.

En apenas siete años, el campo ha pasado de bocetos teóricos a demostraciones físicas de distribución cuántica.



Estado del arte

La computación cuántica distribuida es un campo muy joven.

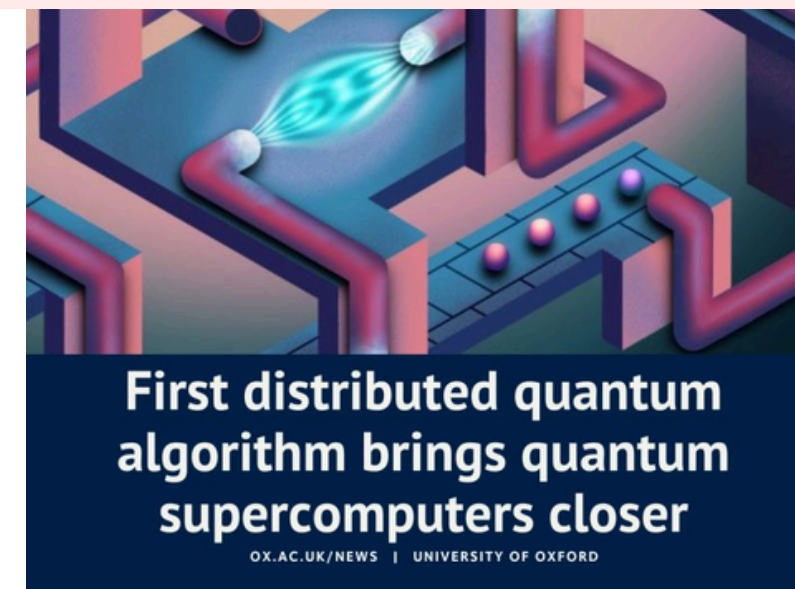
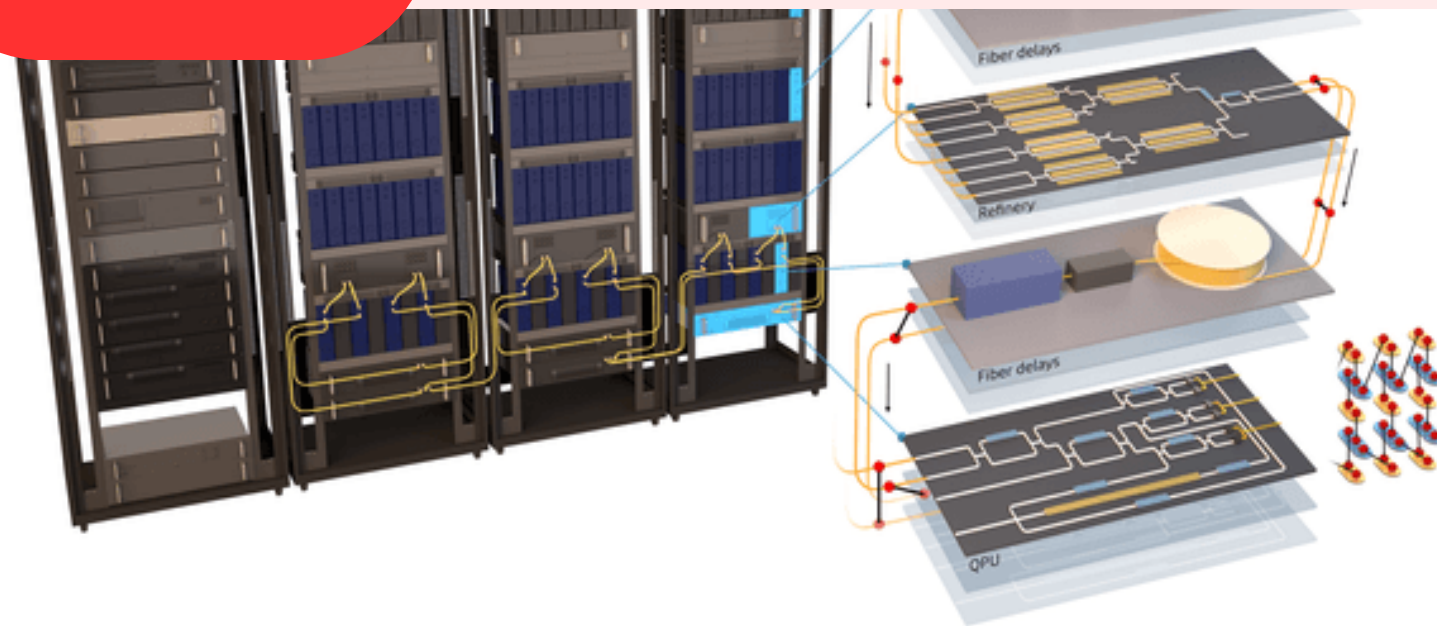
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En apenas siete años se ha pasado de bocetos teóricos a demostraciones físicas de distribución cuántica.



Sistema **accesible** que mantenga **computación** cuántica **razonable** en cada nodo y esté **interconectado** por una **red cuántica programable**.

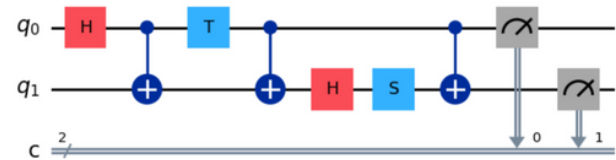


Abstracciones científicas

El problema de mapeo en la Computación Cuántica Distribuida

Carga de trabajo cuántico

Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...

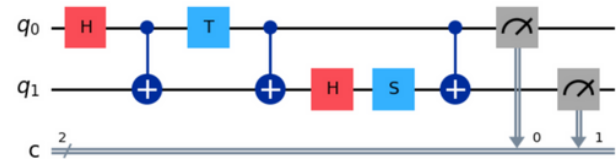


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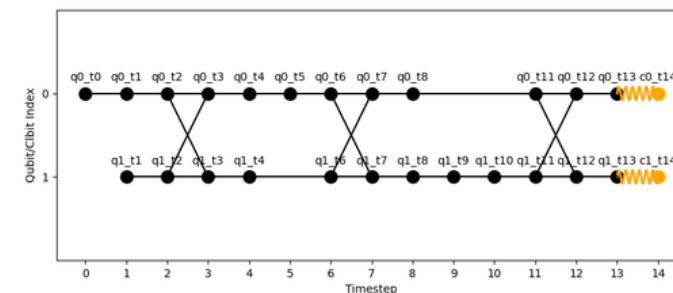
Carga de trabajo cuántico

Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...



Abstracción intermedia

Una representación que captura las dependencias de los datos a lo largo de la computación.



Hypergraph abstraction

POSTER: Distributed Quantum Computing Across Heterogeneous Hardware with Hybrid Dependency Hypergraphs

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Abstract

Distributing quantum computations across heterogeneous devices introduces communication and coordination costs that depend on both quantum and classical operations. We present an architecture-aware approach using Hybrid Dependency Hypergraphs (HDHs) to model space, time, and type dependencies in distributed execution. This poster explores how HDHs support network-level reasoning about communication patterns, and how they can expose cost trade-offs across circuit-based, measurement-based, and quantum walk models. We highlight patterns in HDH structure that inform scheduling, device assignment, and cut placement in heterogeneous quantum clusters.

CCS Concepts

• **Computing methodologies** → **Distributed computing methodologies**; • **Theory of computation** → **Quantum computation theory**; • **Software and its engineering** → *Compilers*.

Keywords

Distributed quantum computing, Quantum compilation, Hybrid Dependency Hypergraphs, Model-agnostic representation

ACM Reference Format:

Maria Gragera Garces, Chris Heunen, and Mahesh Marina . 2025. POSTER: Distributed Quantum Computing Across Heterogeneous Hardware with Hybrid Dependency Hypergraphs. In *ACM SIGPLAN 2025 Conference on*

(MBQC) [10], quantum cellular automata (QCA) [6] or quantum walks (QW) [9]. This narrow focus limits deployment, as different platforms support different computational models and hybrid workflows are central to many protocols. Notably, photonic quantum computers, which natively interface with quantum networks, do not operate within the circuit paradigm [2].

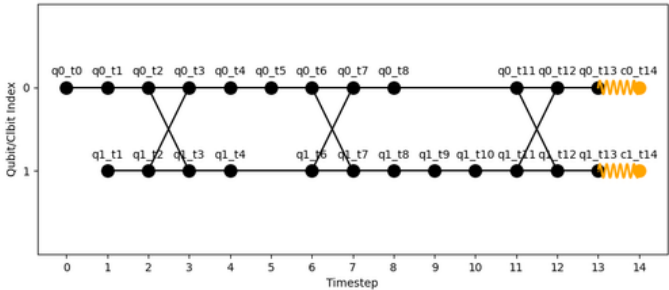
We recently introduced Hybrid Dependency Hypergraphs (HDHs) [7]: a model-agnostic formalism that represents any quantum computation, including classical control, as a typed hypergraph. HDHs capture space, time, and type dependencies, enabling minimum cost, meaning communication, distributions across heterogeneous architectures.

This poster presents HDHs as a foundation for unified, architecture-aware distribution. We present how quantum workloads induce HDHs, and how emerging structural patterns in HDHs could inform efficient distributed implementations.

2 Hybrid Dependency Hypergraphs

HDHs offer a unifying abstraction for distributing quantum computations across different computational models. As a typed hypergraph representation of quantum workloads, they capture both classical and quantum operations within a single structure that mirrors the computational flow:

- **Nodes**: represent intermediate computational states (quantum or classical)



HDHs

HDH

Search docs

Home

An introduction to DQC

A brief introduction to HDHs

Quantum Computational Model mappings to HDHs

Visualizing HDHs

The HDH Database

Partitioning HDHs for distribution

Related publications and information

Home

Welcome to the HDH library

HDH refers to *Hybrid Dependency Hypergraph*, an abstraction developed to enable the partitioning of quantum computations in the context of Distributed Quantum Computing. HDHs are a directed hypergraph based abstraction that encodes the dependencies generated by entangling quantum operations displaying the state transformations performed along the computation. They aim to serve as a unifying abstraction capable of encoding any quantum workload regardless of the computational model it is designed in, that enables all valid partitions of a computation (superseding telegate and teledata abstractions). Furthermore HDHs, as their name implies, also encode classical information enabling the outline of natural classical partitioning points, such as mid-circuit measurements.

You can find an in depth description of HDHs as an abstraction here: [An introduction to HDHs](#). Further explanations of how HDHs are generated from quantum computational models can be found here: [Generation of HDHs from model instructions](#). You can also find a Database with over 2000 HDHs here: [HDH Database](#).

The source code can be found: <https://github.com/grageragarces/HDH>. If you find any bugs or have any proposals for the library we encourage you to open an issue. A guide on how to do this can be found [here](#).

HDHs were originally developed by Maria Gragera Garces, Chris Heunen and Mahesh K. Marina. Publications, posters and talks related to the HDH project can be found here: [Literature](#). The library is currently under MIT License. The development of this library was kindly supported by a [Unitary Fund](#) microgrant, as well as the Engineering and Physical Sciences Research Council (grant number EP/W524384/1), the University of Edinburgh, and [VeriQloud](#).

grageragarces / HDH

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grageragarces Merge pull request #25 from cosenal/fix-quick-start-imports ef516c1 · 1 hour ago 64 Commits

.github/workflows	METIS-based HDH partitioning with evaluation metrics	4 months ago
docs	Update docs with model tables	yesterday
hdh	remove cached files	last week
miscellaneous	remove cached files	last week
tests	remove cached files	last week
.gitignore	Update docs with model tables	yesterday
LICENSE	setup for publishing	4 months ago

About

HDH-based IR for compilation of distributed quantum workloads

[intermediate-representation](#) [distributed-quantum-computing](#) [hdh](#)

Readme

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Activity

6 stars

0 watching

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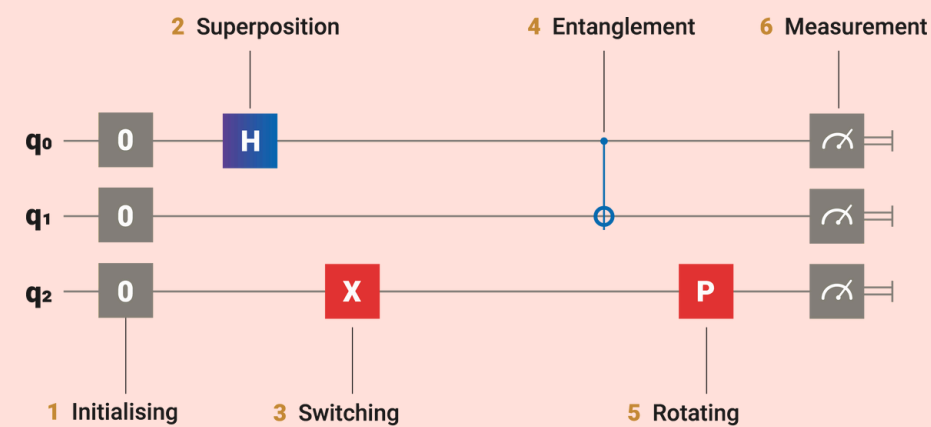
HDHs

Carga de trabajo cuántica

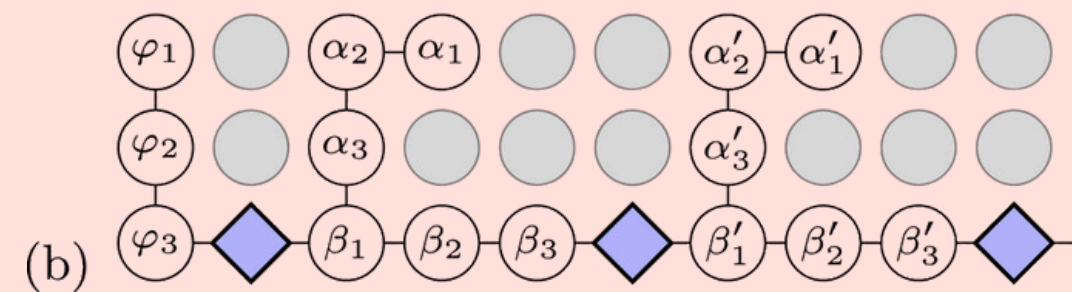
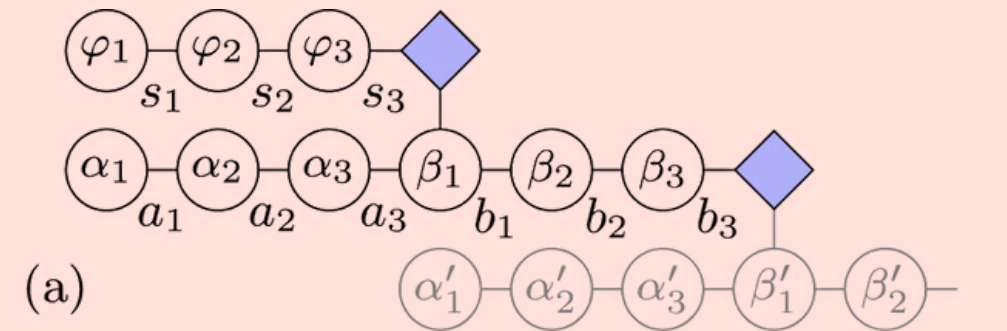
Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...

Existen varios modelos de computación cuántica

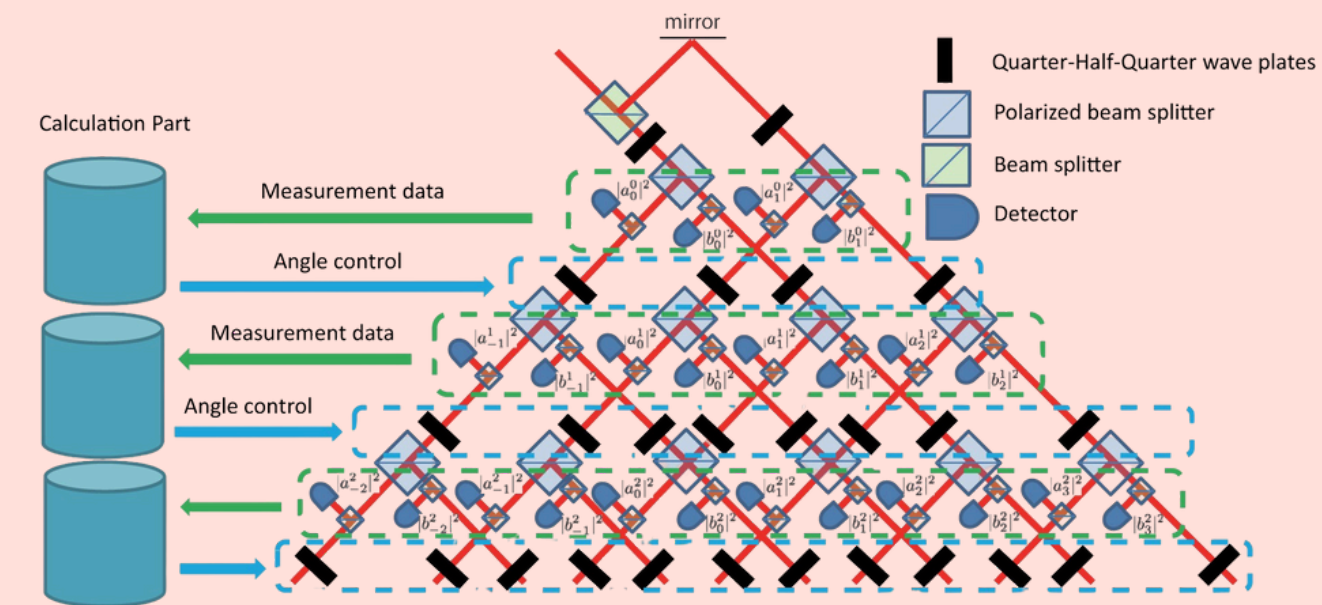
Circuits



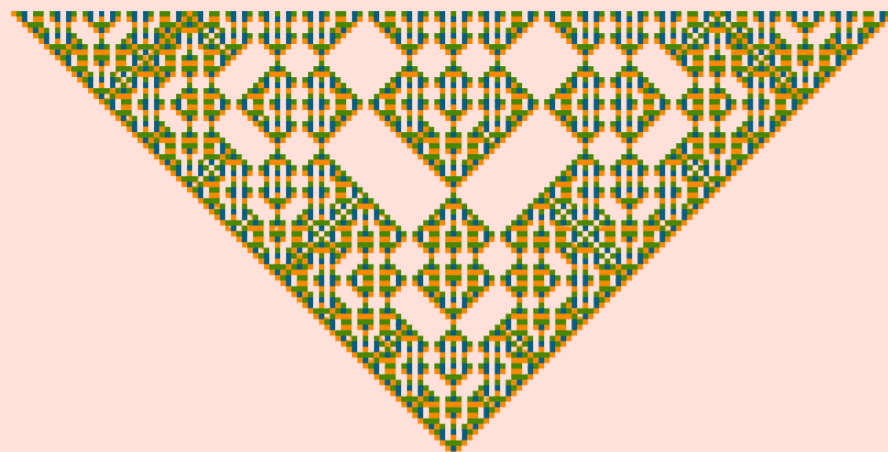
MBQC Patterns



Quantum walks



Quantum cellular automata



HDHs

Abstracción intermedia

Una representación que captura las dependencias de los datos a lo largo de la computación.

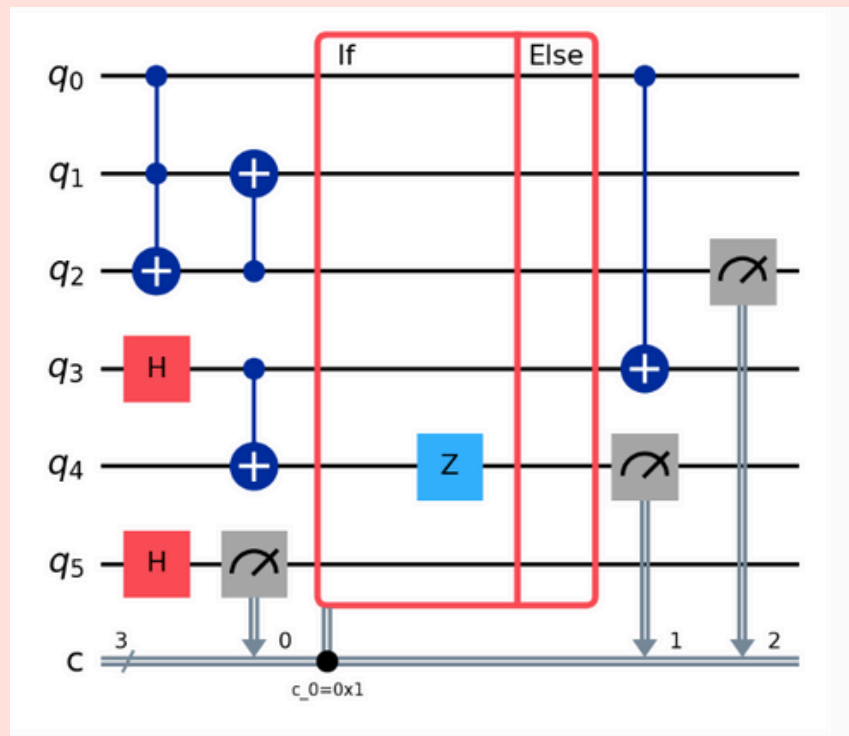
El objetivo es representar todas estas computaciones de manera que podamos evaluar el mínimo cost de su distribución

Symbol	Meaning
●	Classical node
●	Quantum node
~	Classical hyperedge
—	Quantum hyperedge
()	Predicted node
--	Predicted hyperedge

HDHs

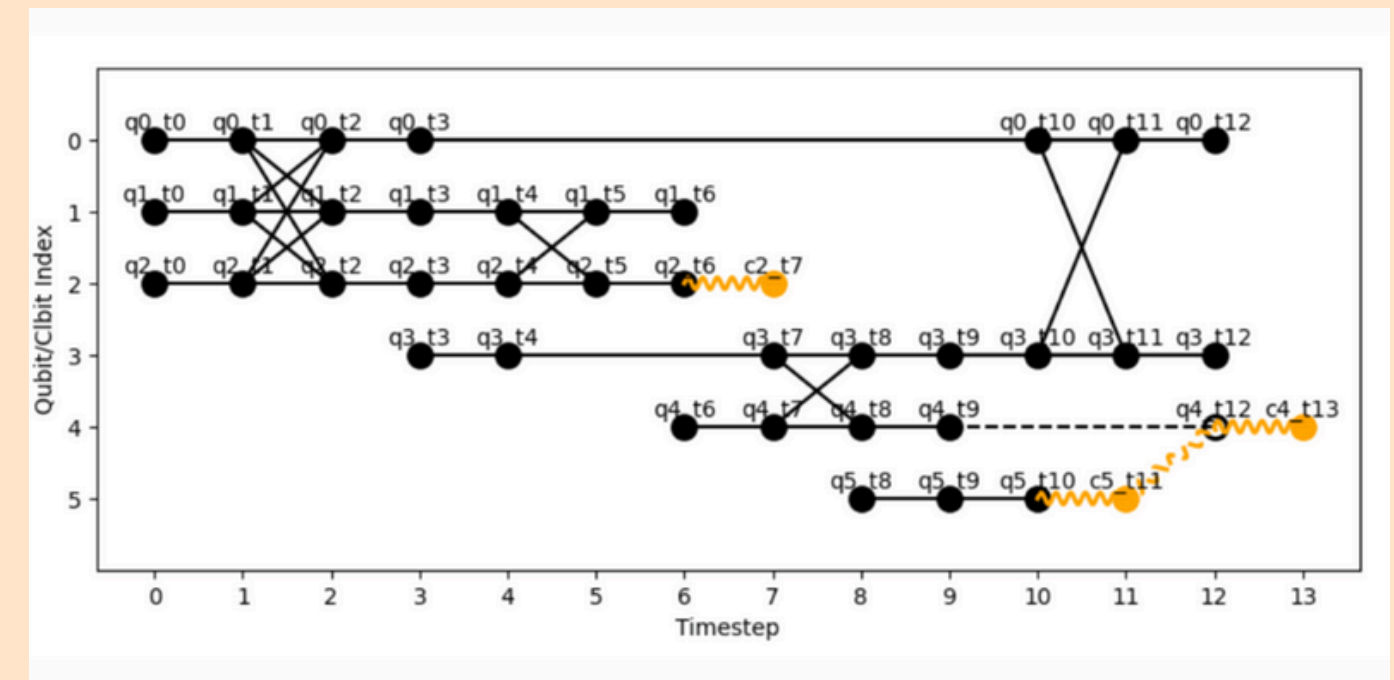
Carga de trabajo cuántico

Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...



Abstracción intermedia

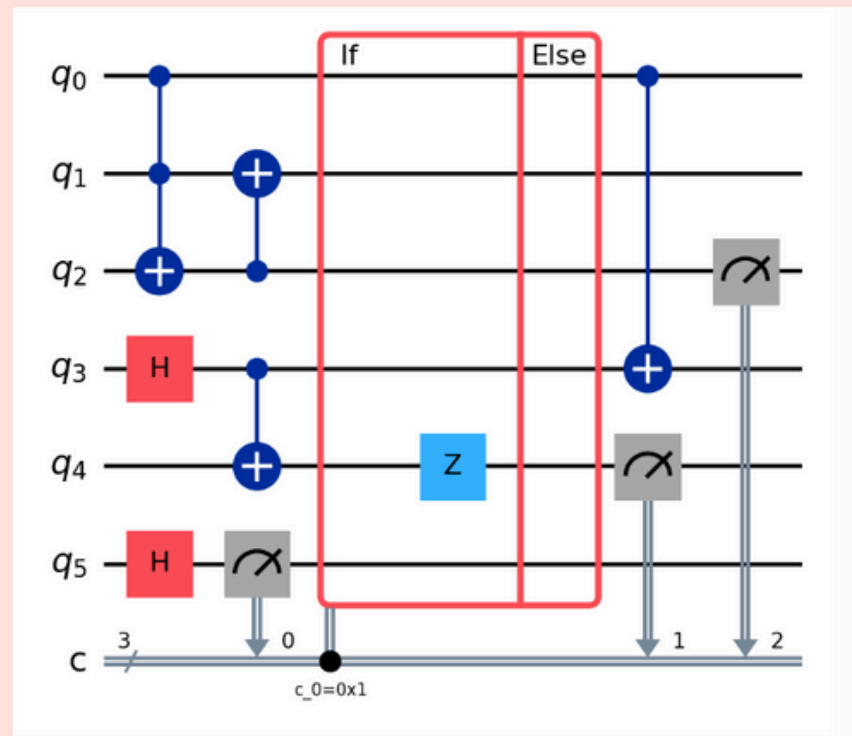
Una representación que captura las dependencias de los datos a lo largo de la computación.



HDHs

Carga de trabajo cuántico

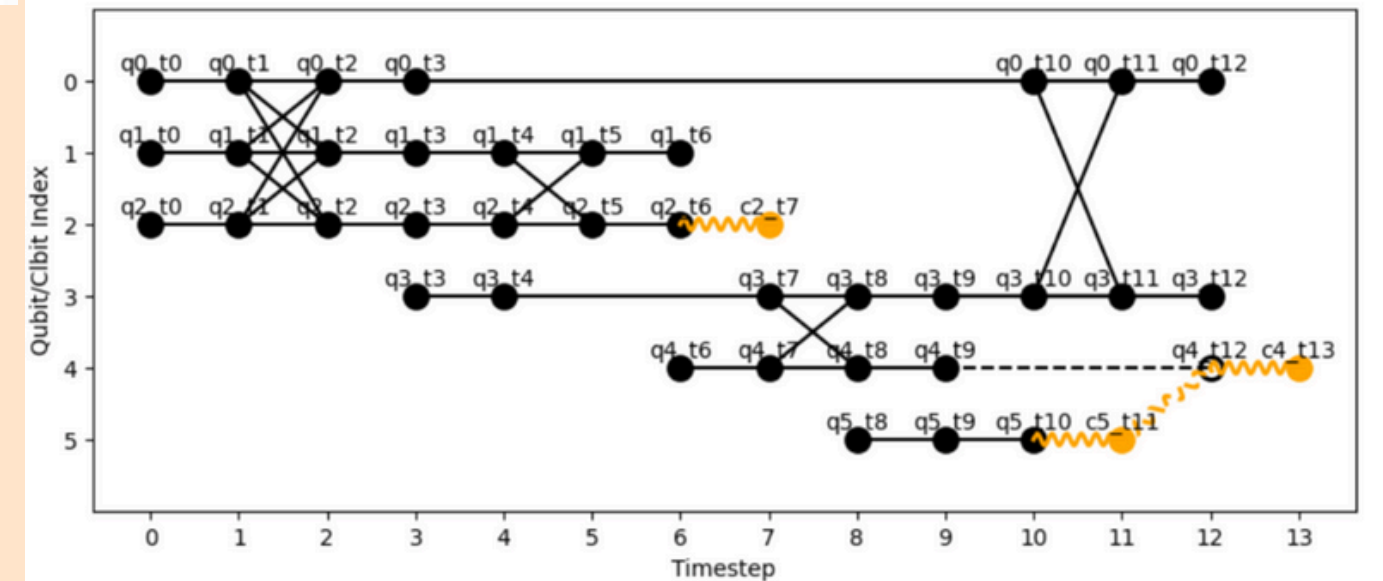
Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation



Motif	Operation
	Qubit initialization
	Single qubit gate
	Two qubit gate
	Three qubit gate
	Measurement

Abstracción intermedia

Abstracción intermedia captura las dependencias de los datos a lo largo de la computación.

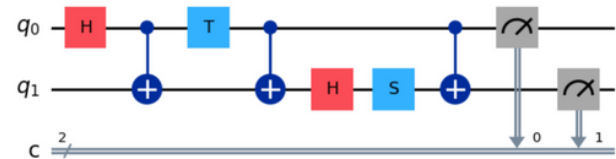


Abstracciones científicas

El problema de mapeo en la Computación Cuántica Distribuida

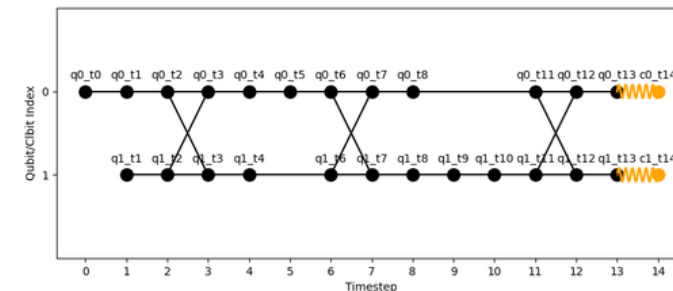
Carga de trabajo cuántico

Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...



Abstracción intermedia

Una representación que captura las dependencias de los datos a lo largo de la computación.



Hypergraph abstraction

Particionamiento

La representación se divide en subcargas de trabajo que pueden calcularse de forma secuencial o en paralelo en una red de dispositivos cuánticos y clásicos.

Hypergraph Partitioning Problem

Partition hypergraph $H = (V, E, c, \omega)$ into k disjoint blocks $\Pi = \{V_1, \dots, V_k\}$ such that:

- blocks V_i are roughly equal-sized:

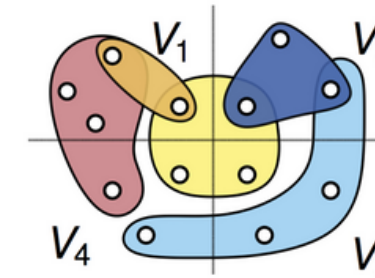
$$c(V_i) \leq (1 + \epsilon) \left\lceil \frac{c(V)}{k} \right\rceil$$

- objective function is minimized:

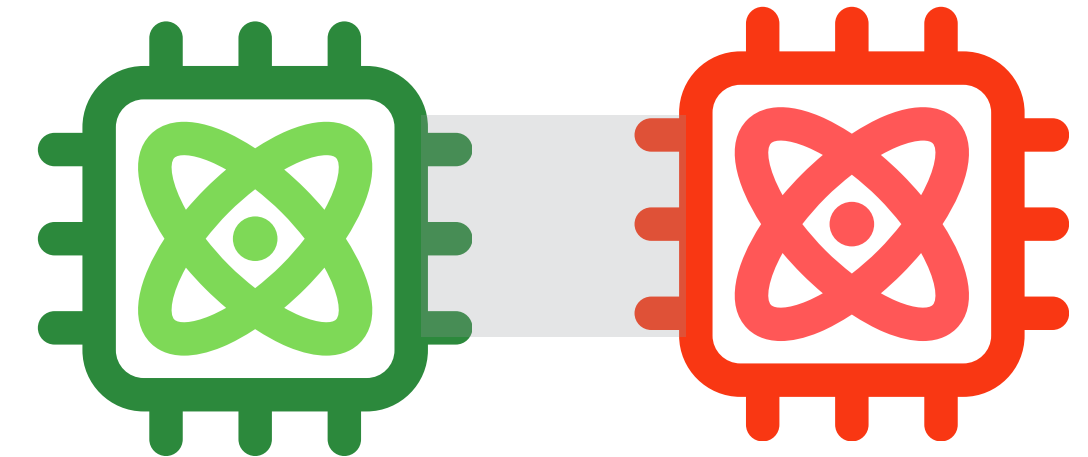
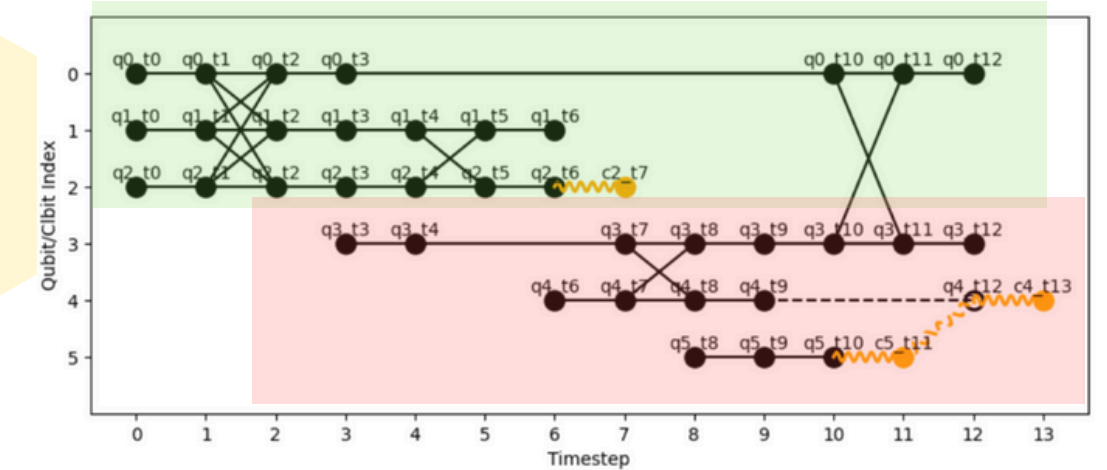
cut-net:
 $\sum_{e \in \text{cut}} \omega(e) = 3$

connectivity:
 $\sum_{e \in \text{cut}} (\lambda - 1) \omega(e) = 6$

blocks connected by net e



kahypar.org



Abstracciones científicas

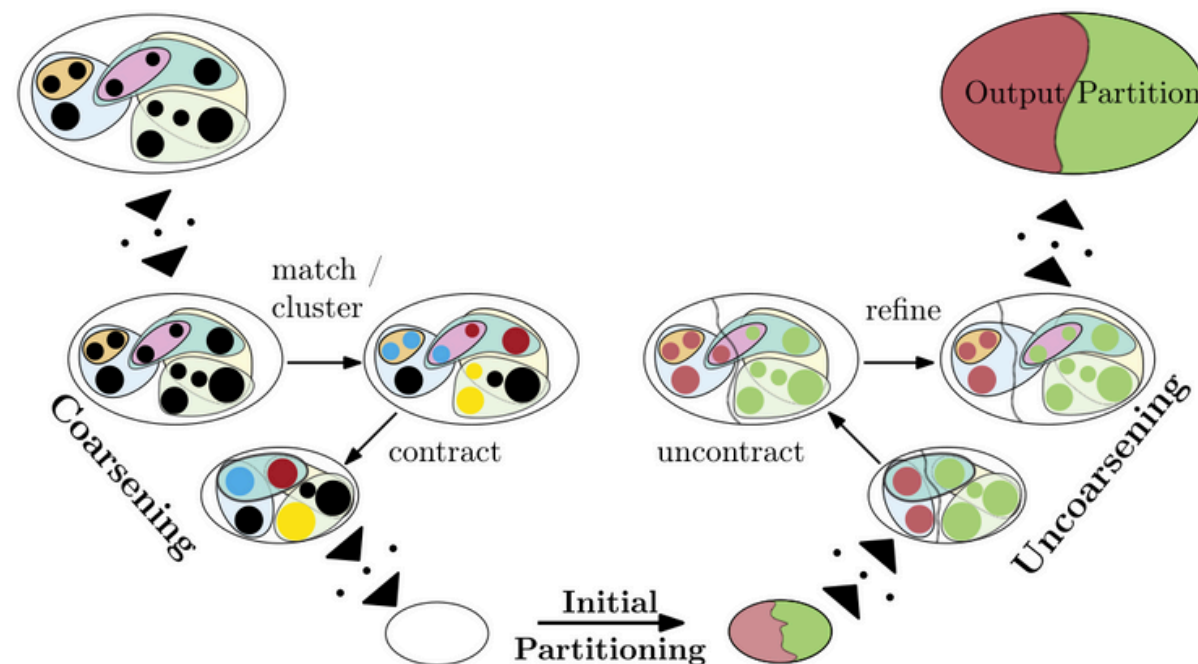
KaHyPar

Fiduccia-Mattheyses

Abstracciones científicas

KaHyPar

1. **Agrandamiento** (Coarsening): los nodos se combinan iterativamente en supernodos, basándose en heurísticas de similitud como la coincidencia de aristas pesadas, reduciendo el tamaño del problema mientras se intenta preservar la estructura esencial del corte.
2. **Particionamiento inicial**: el hipergrafo reducido se particiona utilizando un submétodo (a menudo un particionamiento voraz o el algoritmo FM), bajo el supuesto de que un buen corte en el hipergrafo reducido también será bueno en el gráfico original.
3. **Desagrandamiento y refinamiento** (Un-coarsening and refinement): el hipergrafo se expande gradualmente hasta recuperar su tamaño original, aplicando heurísticas de mejora local en cada etapa para reasignar nodos de frontera y reducir aún más el tamaño del corte.

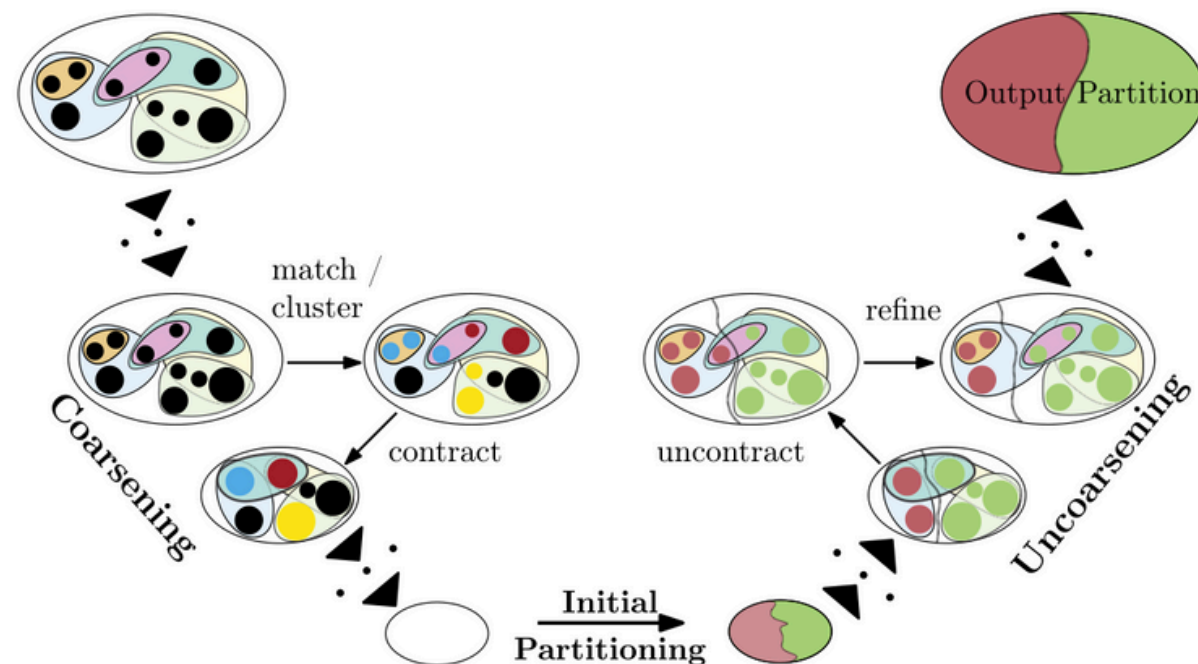


Fiduccia-Mattheyses

Abstracciones científicas

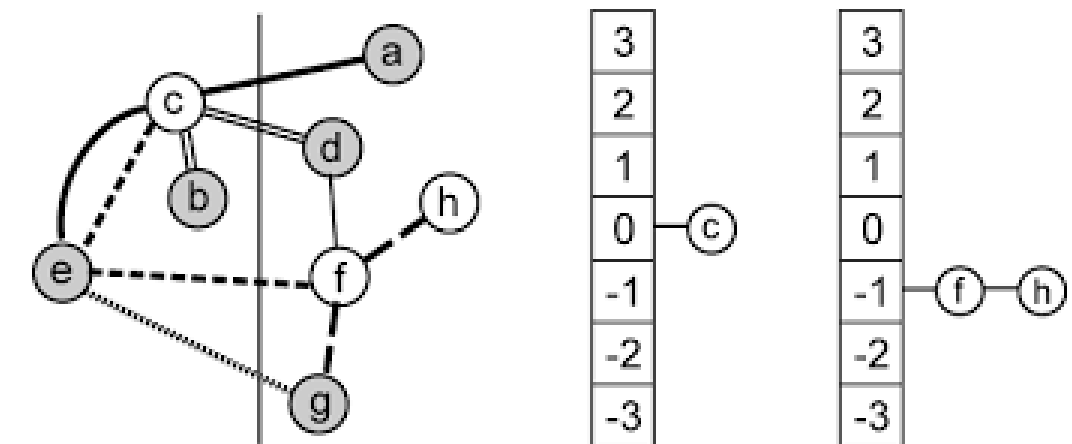
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Fiduccia-Mattheyses

1. **Cálculo de ganancia**: para cada nodo v , se calcula su ganancia
2. **Selección de nodo**: se elige el nodo con la mayor ganancia que pueda moverse sin violar la restricción de equilibrio.
3. **Movimiento y bloqueo**: se mueve el nodo seleccionado a la otra partición y se bloquea para que no pueda volver a moverse durante la pasada actual.
4. **Actualización de vecinos**: se recalculan las ganancias de los nodos vecinos para reflejar la nueva estructura de partición.
5. **Iteración**: se repiten los movimientos de nodos hasta que no sean posibles más movimientos beneficiosos bajo la restricción de equilibrio.



Abstracciones científicas

El problema de mapeo en la Computación Cuántica Distribuida

Carga de trabajo cuántico

Circuit/ MBQC pattern/ Quantum walk/ Hybrid computation /...

Abstracción intermedia

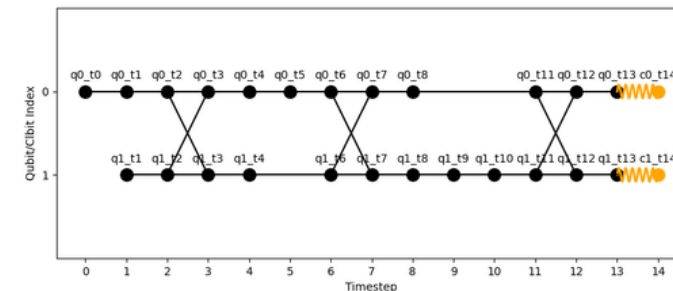
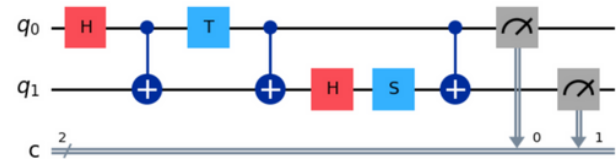
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Particionamiento

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Mapeo a una red híbrida

Las subcargas de trabajo se asignan a una red de dispositivos cuánticos y clásicos que colaboran entre sí. Según el tipo de canal disponible (clásico o cuántico), se utilizan primitivas de comunicación para conectar las particiones.



Hypergraph abstraction

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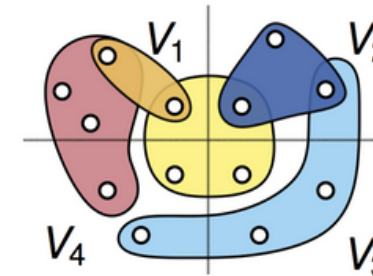
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blocks connected by net e



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?
¿Qué red?

Internet cuántico

El Internet Cuántico es una red global que conectará dispositivos cuánticos (ordenadores, sensores y repetidores) mediante enlaces que distribuyen entrelazamiento.

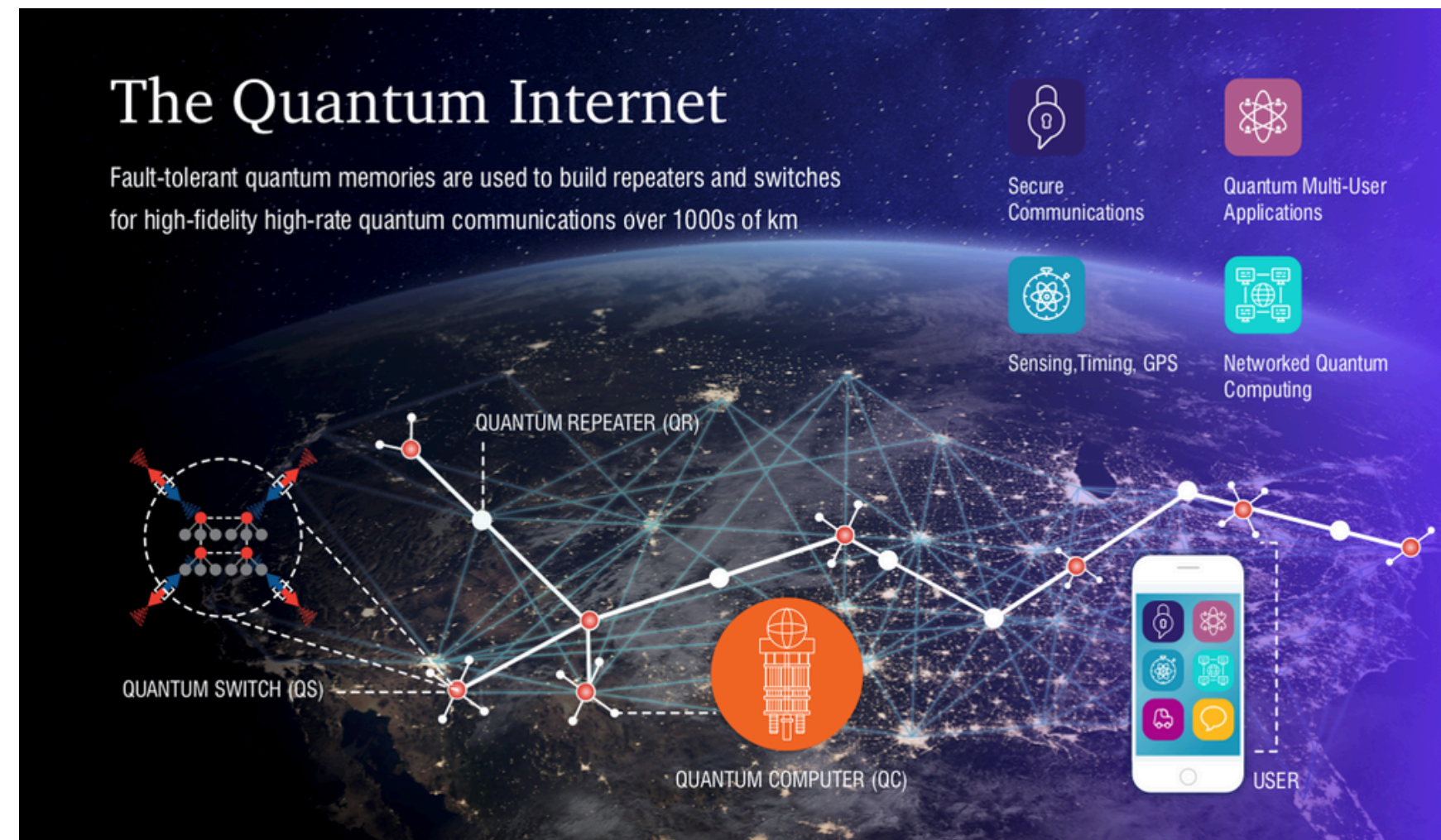
Su objetivo no es transmitir bits clásicos, sino qubits entrelazados, preservando las correlaciones cuánticas a través de grandes distancias.

A diferencia de Internet clásico, esta red permitirá:

- Comunicación absolutamente segura mediante claves cuánticas (QKD).
- Computación cuántica distribuida entre procesadores remotos.
- Sincronización ultraprecisa y nuevos tipos de sensores globales.

Construirlo requiere infraestructura híbrida: canales ópticos, repetidores cuánticos y control clásico coordinado.

Será la base que conecte los nodos del futuro ecosistema cuántico.



IonQ Announces Quantum Networking Breakthrough

Hannah Lanford

Quantum computing and networking startup [IonQ](#), USA, has reportedly attained a significant milestone on the road to scalable quantum networks. The company announced that it successfully demonstrated the frequency conversion of photons from the visible wavelengths used to interface with trapped barium ions into telecom wavelengths on a prototype system.

The company said in a press release accompanying the announcement that this achievement “paves the way for interconnecting quantum computers over vast distances using the current existing fiber optic infrastructure.”



[Image: IonQ]

Toward distributed quantum computing

Photonic quantum computing systems use trapped ion and atomic components that have energy transitions associated with visible wavelengths, which can’t be used in conventional fiber optic links. Converting those visible photons to telecom wavelengths would therefore allow quantum computers to be connected over long distances.

Actores hoy

Executive Platform

Cisco Quantum Labs Announces Software That Networks Quantum Computers Together and Enables New Classical Applications

4 min read

Vijoy Pandey

Xanadu introduces Aurora: world's first scalable, networked and modular quantum computer

January 22, 2025



Quantum transduction and networking for scalable computing applications

Overview

As Google has demonstrated at a global scale, distributing computing resources in a modular fashion offers immense benefits over monolithic computing architectures, including robustness, cost, and performance. This is true at scales ranging from networks within a single computing facility to sharing information across continents. In order to reach maximum potential, quantum technology, such as the [superconducting qubit](#) platforms being developed by Google Quantum AI, will eventually need to operate in a distributed fashion. Within data centers, this has the ability to increase modularity and robustness in design, and decrease requirements for control wiring and cryogenics dramatically. In addition, quantum technology is expected to uniquely benefit from the ability to transmit quantum data collected from the world directly into a quantum computer for processing, in order to learn from quantum data or discover fundamentally new facets of our universe. We expect substantial developments in this area to impact both the design of near-term

Improving interconnects

A fully realized large-scale quantum computer requires millions of physical qubits. However, practical hardware constraints make scaling single chips to these levels prohibitively challenging. As a solution, IBM is developing next-generation interconnects capable of shifting quantum information across multiple chips. This solution provides a modular scalability to reach the required qubits needed to perform error correction. IBM plans to demonstrate these new interconnects—called l-couplers and m-couplers—with proof-of-concept chips called Flamingo and Crossbill, respectively. These couplers are responsible for scaling chips. IBM plans to demonstrate c-couplers by the end of 2026 with a chip called Kookaburra. These are responsible for assisting with error correction.

Scaling quantum processors

While quantum processors relying on qubits used in quantum computing have the potential to massively outperform bit-based processors, current quantum processors can only support a few potential qubits. As research progresses, IBM plans to introduce a quantum system with 200 logical qubits capable of running 100 million quantum gates by 2029, with a goal of 2,000 logical qubits capable of running 1 billion gates by 2033.

Scaling quantum hardware

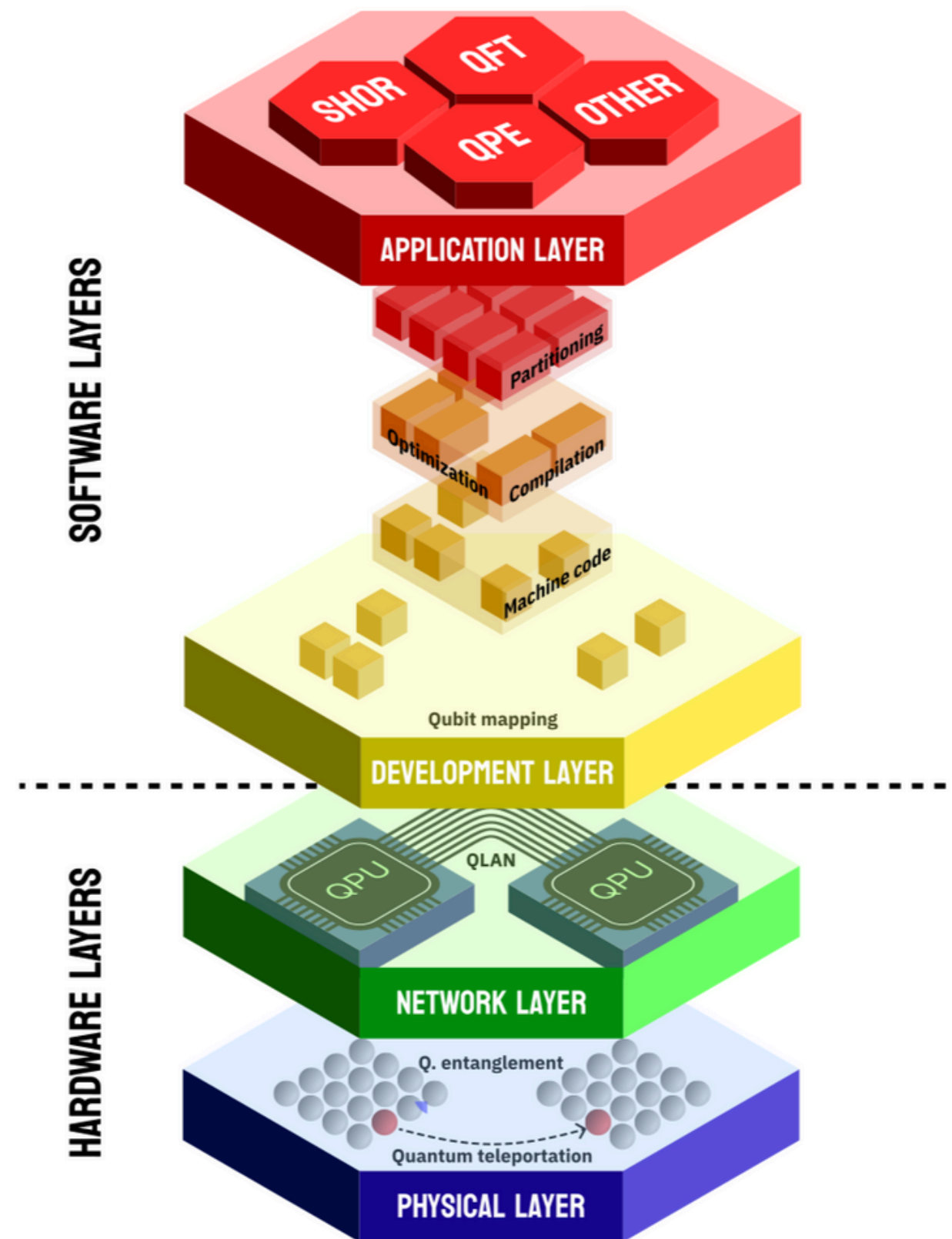
The Quantum Decade

Prepare for the most significant computing revolution in 60 years. Read The Quantum Decade to find out how you, too, can be quantum ready – and how this bleeding-edge technology can help you and your business thrive the moment quantum computers come of age. Because that moment is closer than you think.

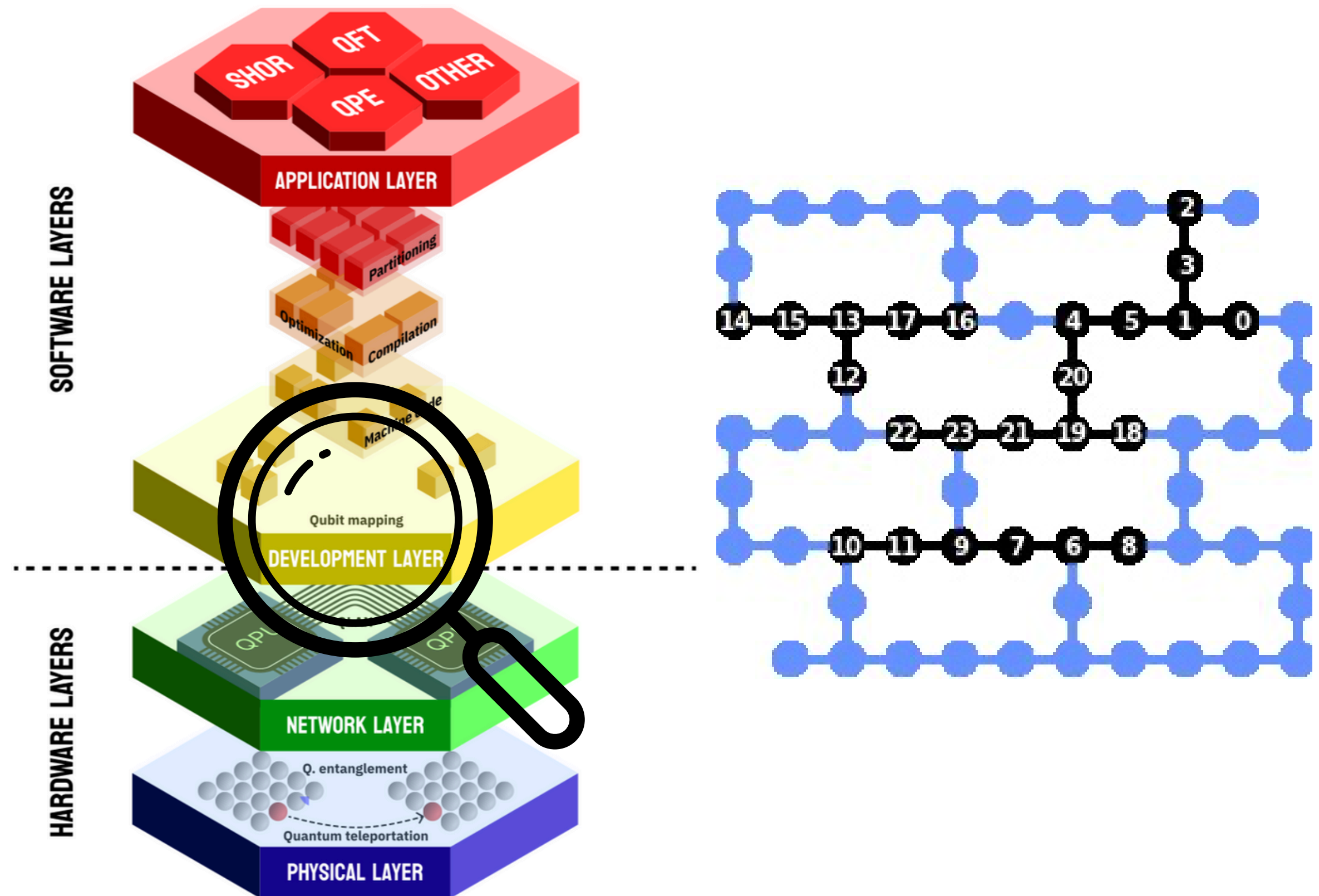
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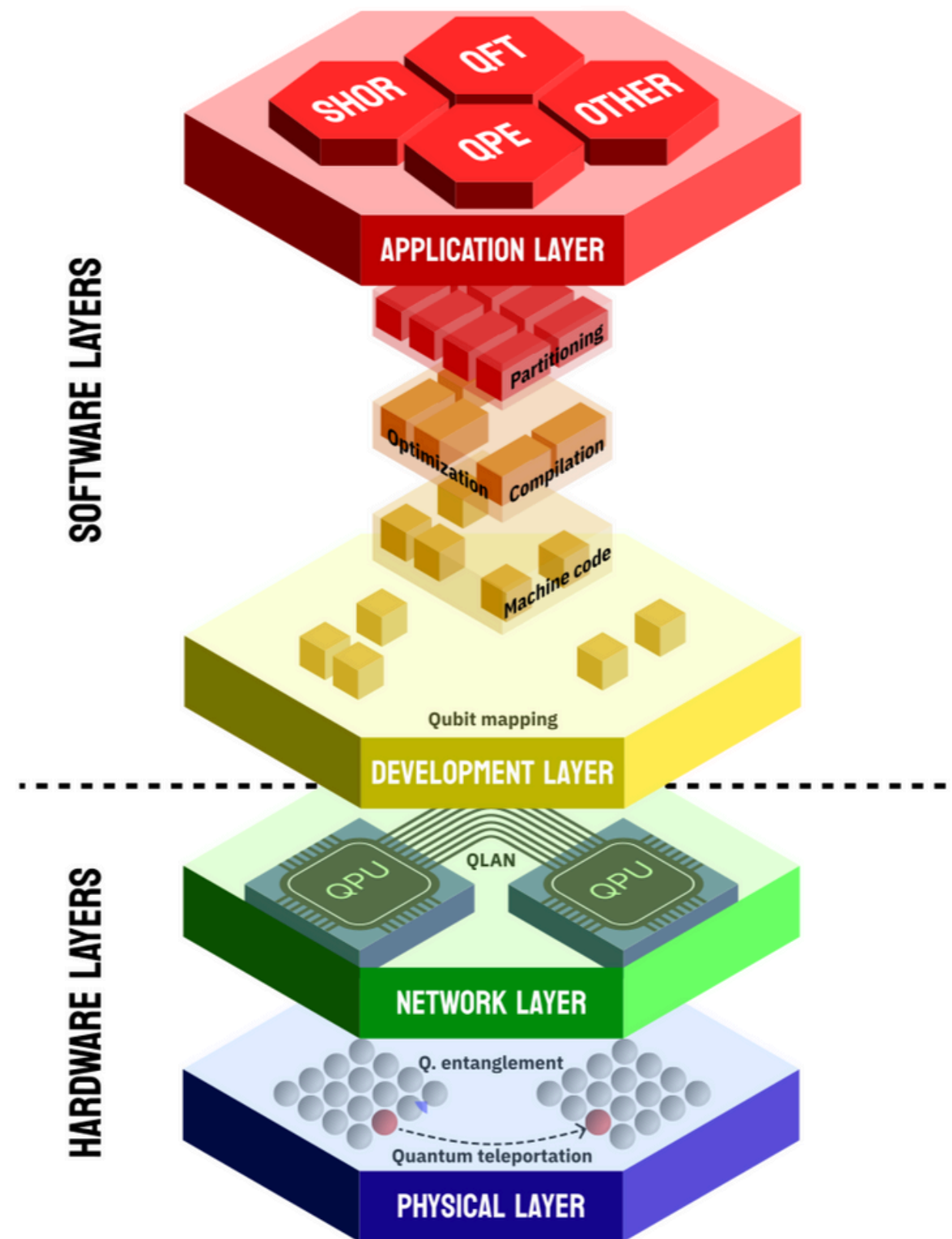
Mucho que construir por delante



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Qiskit addon: circuit cutting

Stars 93

Qiskit addons are a collection of modular tools for building utility-scale workloads powered by Qiskit.

This package implements circuit cutting. In this technique, a handful of gates and/or wires are cut, resulting in smaller circuits that are better suited for execution on hardware. The result of the original circuit can then be reconstructed; however, the trade-off is that the overall number of shots must be increased by a factor exponential in the number of cuts.

For a more detailed discussion on circuit cutting, check out our technical guide.

We follow semantic versioning and are guided by the principles in Qiskit's deprecation policy. We may occasionally make breaking changes in order to improve the user experience. When possible, we will keep old interfaces and mark them as deprecated, as long as they can co-exist with the new ones. Each substantial improvement, breaking change, or deprecation will be documented in the Release Notes.

Note

This package was known as the Circuit Knitting Toolbox prior to September 2024.

Citing this project

If you use this package in your research, please cite it according to [CITATION.bib](#) file included in this repository:

```
@misc{qiskit-addon-cutting,
  author = {
    Agata M. Bra\{'n}czyk
    and Almudena {Carrera Vazquez}
    and Daniel J. Egger
    and Bryce Fuller
    and Julien Garon
```

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Publicaciones

Automated distribution of quantum circuits

Pablo Andrés-Martínez^{*} and Chris Heunen[†]
University of Edinburgh
(Dated: November 27, 2018)

Quantum algorithms are usually monolithic circuits, becoming large at modest input size. Near-term quantum architectures can only execute small circuits. We develop automated methods to distribute quantum circuits over multiple agents, minimising quantum communication between them. We reduce the problem to hypergraph partitioning, which is NP-hard but has advanced heuristic solvers. Our implementation is evaluated on five quantum circuits. Four are amenable to distribution. The distribution cost is more than halved when compared to a naive approach.

PACS numbers: 03.67.Ac; 03.67.Lx

Quantum computation [1–3] leverages the laws of quantum mechanics to design physical systems capable of outperforming devices based on the laws of classical mechanics [4–6]. Over the past couple of decades, this idea has rapidly developed from theoretical results into actual quantum technology [7–9].

Although there are other approaches [10], the dominant way to present a quantum algorithm is as a quantum circuit [11]: a description of how quantum devices, chosen from a fixed finite set, are applied to different parts of the input system; see Figure 1 for an example. Each of the ‘wires’ that quantum devices act upon typically consist of a two level quantum system called a quantum

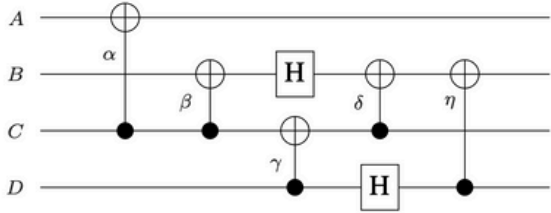
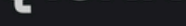


FIG. 1. Example quantum circuit, applying Hadamard gates H and CNOT gates $\alpha, \beta, \gamma, \delta$ to four input qubits A, B, C, D .

Simulaciones



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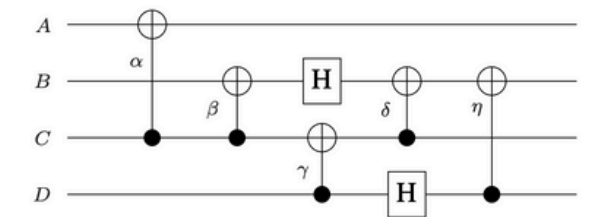


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Open Source

The screenshot shows the GitHub repository page for **grageragarcas / HDH**. The **Issues** tab is highlighted with a red circle. The repository is public and has 10 issues, 4 forks, and 7 stars. The main branch is **main** with 4 branches and 7 tags. The repository description is **HDH-based IR for compilation of distributed quantum workloads**. The file list shows **.github/workflows**, **docs**, **hdh**, and **miscellaneous** folders.

Conclusion

La Computación Cuántica Distribuida permitirá:

- 1. Escalar sin depender de un único dispositivo.**
- 2. Interconectar tecnologías heterogéneas.**
- 3. Optimizar recursos cuánticos y clásicos de forma cooperativa.**

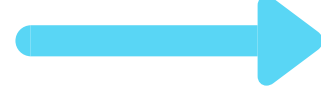
La próxima frontera no es un ordenador más grande, sino una red más inteligente.



Gracias!



LinkedIn



Bluesky



<https://grageragarces.github.io>

GRACIAS



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